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Assignment 4

Q1 let total no. of units produced by product A be x

let total no. of units produced by Product B be y

Profit of product A per unit = ₹46

Profit of product B per unit = ₹30

=> $\boxed{Z = 40x + 30y}$

Maximize

=> Product A takes 2 hours & B takes 1 hour for Machine 1

Maximum available time = 100 hours

Equation: $\boxed{2x + 1y \leq 100}$

Product A takes 1 hours & B takes 2 hours for Machine 2

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2) Maximum available time = 80 hours

Equation: $\boxed{1x + 2y \le 80}$

⇒ $\boxed{x \ge 0, \; y \ge 0}$

Step → 1
$2x + y = 100$ — (i) Eq
$x + 2y = 80$ — (ii) — Eq
$x = 0$
$y = 0$

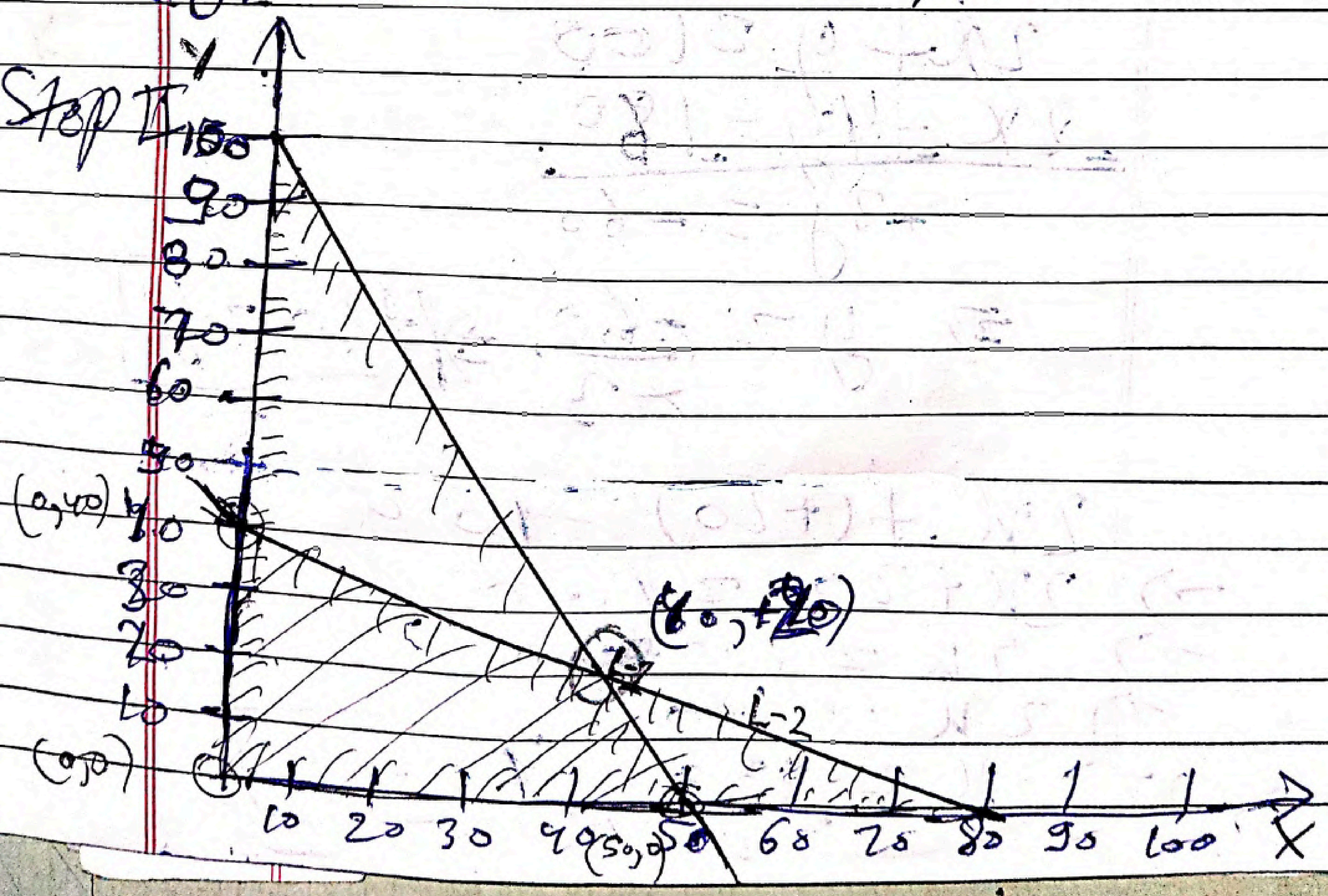
x	0	80
y	100	0

— (i) — Eq

x	0	80
y	40	0

— (ii) — Eq

Step II



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$Z = 40x + 30y$

putting corner point values

$\Rightarrow (0, 40)$ in Eq

$\Rightarrow 40 \times 0 + 30 \times 40$

$\Rightarrow 1200 = Z$

$\Rightarrow (0, 0)$ in Eq

$\Rightarrow 40 \times 0 + 30 \times 0 \Rightarrow 0 = Z$

$\Rightarrow (50, 0)$ in Eq

$\Rightarrow 50 \times 40 + 30 \times 0 \Rightarrow 2000 = Z$

$\cancel{2x} + y = 100$

$\underline{-\cancel{2x} + 4y = 180}$

$-3y = -60$

$\Rightarrow y = \frac{-60}{-3} \Rightarrow \boxed{20 = y}$

$2x + (20) = 100$

$\Rightarrow 2x + 20 = 100$

$\Rightarrow 2x = 100 - 20$

$\Rightarrow 2x = 80$

$\Rightarrow x = \frac{80}{2} \Rightarrow \boxed{40 = x}$

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~~(60, 40)~~
(40, 20)

putting in Eq

$40 \times 40 + 30 \times 20$
$\Rightarrow 1600 + 600 \Rightarrow 2200$

Corner point	$Z = 40x + 30y$	
(0, 40)	1200	
(0, 0)	0	
(50, 0)	2000	
(40, 20)	2200	← Maximum

So maximum value is 2200 at (40, 20)

Q2 $Z = 5x + 3y$
$2x + y \le 10$
$x + y \le 6$
$x, y \ge 0$

step-1 $2x + y = 10$ — (i) — Eq
$x + y = 6$ — (ii) — Eq
$x = 0$
$y = 0$

x	0	5
y	10	0

— (i) Eq

x	0	6
y	6	0

— (ii) Eq

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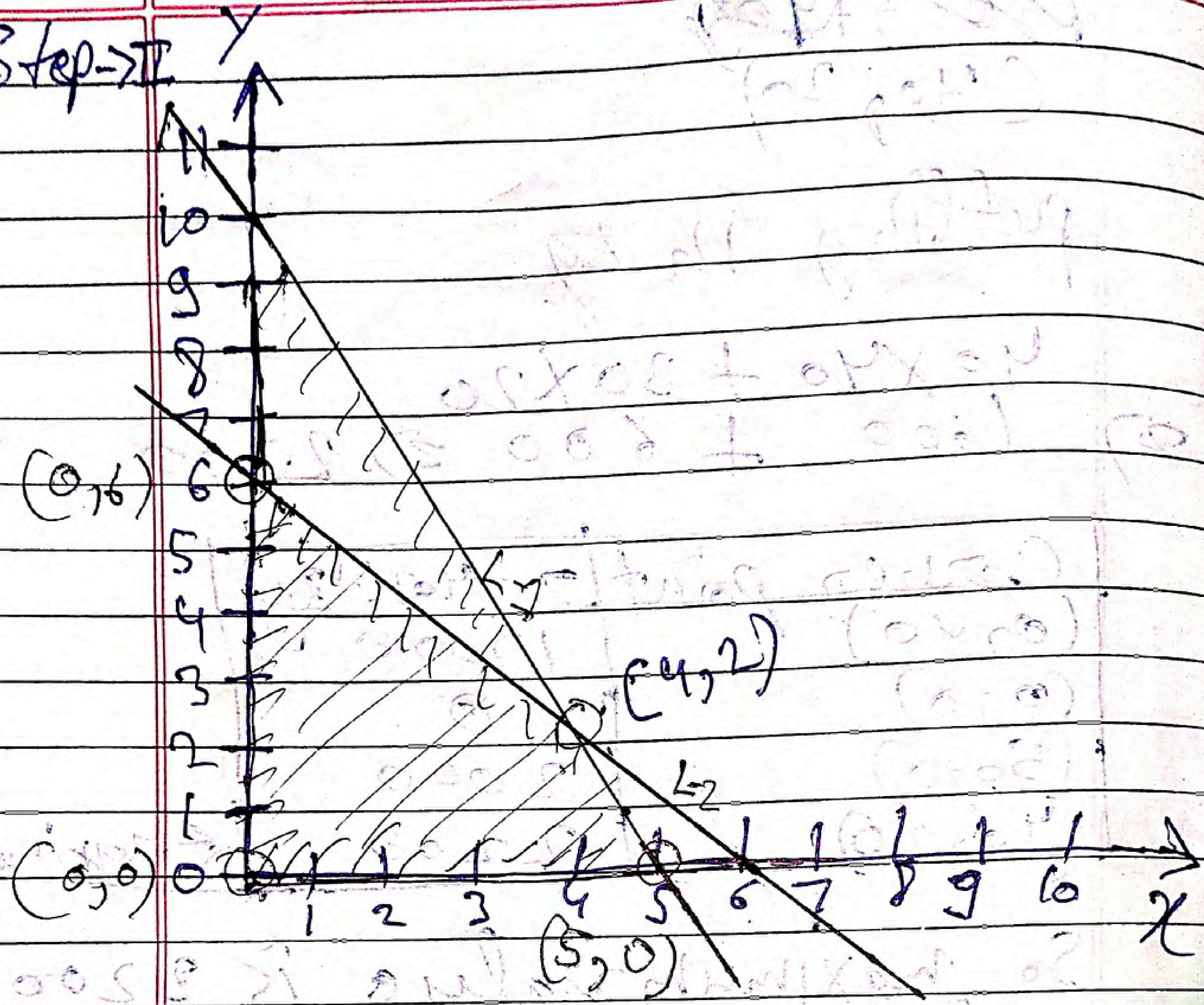
Step → II



$2x + y = 10$
$2x + 2y = 12$

$-y = -2$

$\boxed{y = 2}$

$2x + 2 = 10$
$\Rightarrow 2x = 10 - 2$
$\Rightarrow x = \frac{8}{2} \Rightarrow \boxed{4 = x}$

$Z = 5x + 3y$

putting corner values in eq

$(0,6)$

$\Rightarrow 5\times0 + 3\times6 \Rightarrow 18$

$(0,0)$

$\Rightarrow 5\times0 + 3\times0 \Rightarrow 0$

$(4,2)$

$\Rightarrow 5\times4 + 3\times2 \Rightarrow 26$

$(5,0)$

$\Rightarrow 5\times5 + 3\times0 \Rightarrow 25$

Corner point	$Z = 5x+3y$	
$(0,6)$	18	
$(0,0)$	0	
$(4,2)$	26	← Maximum
$(5,0)$	25	

So, maximum value is 26 at $(4,2)$.

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Q3 $z = 2x + 3y$

$x + 2y \geq 8$
$3x + y \geq 9$

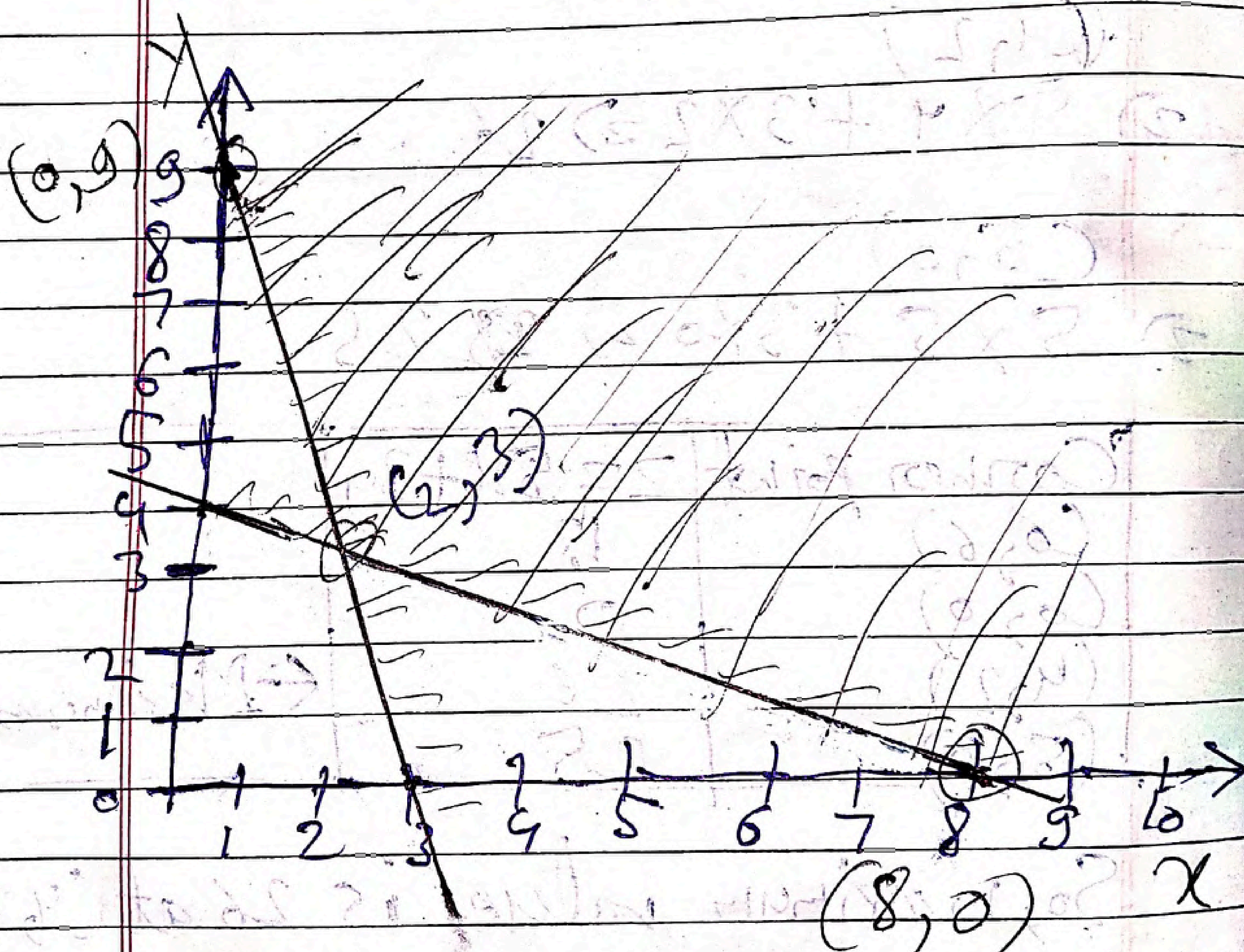
Step 1 => $x + 2y = 8$ — (i) eq
$3x + y = 9$ — (ii) eq

x	0	8
y	4	0

— (i) eq

x	0	3
y	9	0

— (ii) eq



$3x + y = 9$
$3x + 6y = 24$
$\quad - \quad\;\; - \quad\;\; -$

$-5y = -15$

$\boxed{y = 3}$

$\cancel{3x + 18 = 9}$

$\cancel{\Rightarrow 3x = 9 - 18}$

$\cancel{\Rightarrow 3x = -6}$

$\cancel{\Rightarrow x = \frac{-6}{3}}$

$\Rightarrow 3x + 3 = 9$

$\Rightarrow 3x = 9 - 3$

$\Rightarrow 3x = 6$

$\Rightarrow x = \frac{6}{3} \Rightarrow \boxed{2 = x}$

$Z = 2x + 3y$

putting values of corners in eq

$(0, 9)$

$\Rightarrow 2 \times 0 + 3 \times 9 = Z$

$\Rightarrow 27 = Z$

$(2, 3)$

$\Rightarrow 2 \times 2 + 3 \times 3 = Z$

$\Rightarrow 13 = Z$

$(8, 0)$

$\Rightarrow 2 \times 8 + 3 \times 0 = Z$

$\Rightarrow 16 = Z$

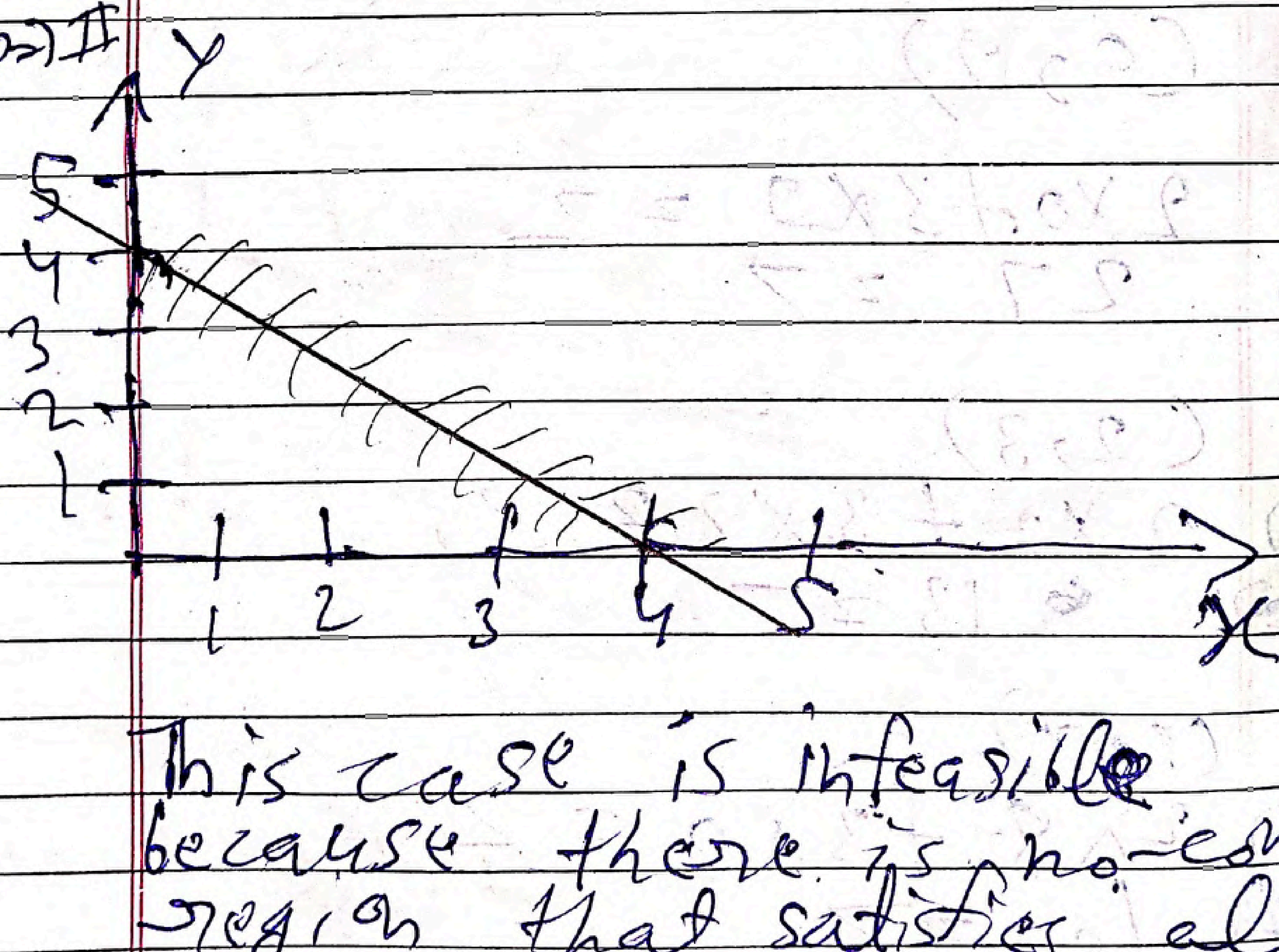
Corner point	$z = 2x+3y$	
(0,9)	27	
(2,3)	13	
(8,0)	16	← Minimum

Q4 $z = x+y$
$x+y \geq 4$
$x+y \leq 4$

Step 1 → $x+y = 4$
$x+y = 4$

x	0	4
y	4	0

Step II



This case is infeasible because there is no common region that satisfies all constants.

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Q5 $Z = 3x + 5y$

$x + 2y \le 8$

$3x + 2y \le 12$

$x, y \ge 0$

$\Rightarrow$ $x + 2y + s_1 + 0s_2 = 8$

$3x + 2y + 0s_1 + s_2 = 12$

$x = 0,\ y = 0$

Max $z = 3x + 5y + 0s_1 + 0s_2$

When $x = 0$ & $y = 0$

$s_1 = 8$

$s_2 = 12$

		Cj	3	5	0	0	
Cj	B.V	Sol var	x	y	S1	S2	Ratio
0	S1	8	1	2	1	0	$\frac{8}{2} = 4$ ←Min
0	S2	12	3	2	0	1	$\frac{12}{2} = 6$
	Zj	0	0	0	0	0	
	Cj − Zj		3	5	0	0	
				↑ Max			

		Cj	3	5	0	0	
Cj	B.V	Sol var	x	y	S1	S2	Ratio
5	y	$\frac{8}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{8}{2}/\frac{1}{2} = 8$
0	S2	4	2	0	−1	1	$\frac{4}{2} = 2$ ←Min
	Zj	20	$\frac{5}{2}$	5	$\frac{5}{2}$	0	
	Cj − Zj		$\frac{1}{2}$	0	$-\frac{5}{2}$	0	
			↑ Max				

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		C_j	3	5	0	0	
C_B	B.V	Sol var	x	y	S_1	S_2	Ratio
5	y	3	0	1	$\frac{3}{4}$	$-\frac{1}{4}$	
3	x	2	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	
	Z_j	21	3	5	$\frac{9}{4}$	$\frac{1}{4}$	
	$C_j - Z_j$		0	0	$-\frac{9}{4}$	$-\frac{1}{4}$	

$y = 3$
$x = 2$

put x & y in eq $\Rightarrow 3x + 5y$
$\Rightarrow 3 \times 2 + 5 \times 3$
$\Rightarrow 6 + 15 \Rightarrow 21$ (Ans).

Ans $\Rightarrow$ Maximum value is 21

Q 6 Solve using Simplex Method:
Maximize
subject to: $z = 2x + y$
$x + y \leq 4$
$x + 3y \leq 6$

$\Rightarrow$ $x + y + S_1 + 0S_2 = 4$
$x + 3y + 0S_1 + S_2 = 6$

Max $Z = 2x + y + 0S_1 + 0S_2$

When $x = 0$ & $y = 0$
$S_1 = 4$
$S_2 = 6$

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		C_j	2	1	0	0	
C_j	B.V	Sol var	x	y	S1	S2	Ratio
0	S1	4	1	1	1	0	4 ← Min
0	S2	6	1	3	0	1	6
	Z_j	0	0	0	0	0	
	$C_j - Z_j$		2	1	0	0	
			↑ Max				

C_j	B.V	Sol var	x	y	S1	S2	Ratio
2	x	4	1	1	1	0	
0	S2	2	0	2	-1	1	
	Z_j	8	2	2	2	0	
	$C_j - Z_j$		0	-1	-2	0	

$x = 4$ & $y = 0$ (we take 0 because y is not in B.V)

$z = 2 \times 4 + 0 \Rightarrow 8$ (A)

Q7 Write the dual of the following LP:

Maximize $z = 4x + 3y$

Subject to: $2x + y \leq 10$

$x + 2y \leq 8$

7) Minimize $z = 10y_1 + 8y_2$

$2y_1 + y_2 \geq 4$

$y_1 + 2y_2 \geq 3$

$y_1, y_2 \geq 0$

Q8

	D1	D2	D3	Supply
S1	4	6	8	20
S2	5	3	7	30
S3	6	5	4	25
Demand	25	35	15	

	D1	D2	D3	Supply
S1	4 $\boxed{20}$	6	8	20
S2	5 $\boxed{5}$	3 $\boxed{25}$	7	30
S3	6	5 $\boxed{10}$	4 $\boxed{15}$	25
D	25	35	15	75

Supply = 20+30+25 = 75
Demand = 25+35+15 = 75
Supply = Demand

Optimal Solution = $4\times20 + 5\times5 + 3\times25 + 5\times10 + 4\times15 \Rightarrow 290$ Rs.

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Q9 LCM

	D1	D2	D3	supply
S1	4	6	8	20
S2	5	3	7	30
S3	6	5	4	25
	25	35	15	

	D1	D2	D3	Supply
S1	4 $\boxed{20}$	6	8	20
S2	5	3 $\boxed{30}$	7	30
S3	6 $\boxed{5}$	5 $\boxed{5}$	4 $\boxed{15}$	25
	25	35	15	75

supply = 20+30+25 = 75
Demand = 25+35+15 = 75
supply = Demand

Optimal Solution
4×20 + 3×30 + 6×5 + 5×5 + 4×15
=> 285 (A)

Q10 VAM

	D1	D2	D3	Supply
S1	4	6	8	20
S2	5	3	7	30
S3	6	5	4	25
D	25	35	15	75

Supply = 20+30+25 = 75
Demand = 25+35+15 = 75
Supply = Demand

	D1	D2	D3	Supply	P1	P2	P3	P4
S1	4 $\boxed{20}$	6	8	20	2	2	2	–
S2	5	3 $\boxed{30}$	7	30	2	–	–	–
S3	6 $\boxed{5}$	5 $\boxed{5}$	4 $\boxed{15}$	25	1	1	1	1
D	25	35	15	75				

P1	1	2	3
P2	2	1	4
P3	2	1	–
P4	6	5	–

Optimal Solution
⇒ 4×20 + 3×30 + 6×5 + 5×5 + 4×15
⇒ 285 (Ans).

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Q11

	D1	D2	D3	Supply
S1	4 $\boxed{20}$	6	8	20
S2	5	3 $\boxed{30}$	7	30
S3	6 $\boxed{5}$	5 $\boxed{5}$	4 $\boxed{15}$	25
D	25	35	15	75

$u_1 = -2$

$u_2 = -4$

$u_3 = 0$

$v_1 = 6 \quad v_2 = 5 \quad v_3 = 4$

$C_{ij} = u_i + v_j$

$C_{32} = u_3 + V_2$

$5 = 0 + v_2 \Rightarrow 5 = v_2$

$C_{33} = u_3 + V_3$

$\Rightarrow 4 = 0 + v_3 \Rightarrow 4 = V_3$

$C_{31} = u_3 + v_1$

$\Rightarrow 6 = 0 + v_1 \Rightarrow 6 = v_1$

$C_{22} = u_2 + v_2$

$\Rightarrow 3 = u_2 + 7 \Rightarrow -4 = u_2$

$C_{11} = u_1 + V_1$

$\Rightarrow 4 = u_1 + 6 \Rightarrow -2 = u_1$

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$\Delta_{ij} = C_{ij} - (u_i + v_j)$

$\Delta_{12} = C_{12} - (u_1 + v_2)$

$\Rightarrow \Delta_{12} = 6 - (-2 + 5)$
$\Rightarrow 6 + 3 \Rightarrow 9 = \Delta_{12}$

$\Delta_{13} \Rightarrow C_{13} - (u_1 + v_3)$

$\Delta_{13} \Rightarrow 8 - (-2 + 4)$
$\Rightarrow 8 - 2 \Rightarrow 6 = \Delta_{13}$

$\Delta_{21} = C_{21} - (u_2 + v_1)$
$\Rightarrow 5 - (-4 + 6)$
$\Rightarrow 5 - 2 = 3 \Rightarrow \Delta_{21}$

$\Delta_{23} = C_{23} - (u_2 + v_3)$
$\Rightarrow 7 - (-4 + 4)$
$\Rightarrow 7 - 0 \Rightarrow 7 = \Delta_{23}$

Hence, all Δ_{ij} are +ve, the solution is optimal

Q 13 Solve completely using:

VAM
MoDI

Supply = 20 + 30 + 25 ⇒ 75
Demand = 10 + 25 + 40 ⇒ 75

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	D1	D2	D3	Supply	P1
					~~2-3=4~~
S1	2	3	1 $\boxed{20}$	~~20~~ 0	3-2=4
S2	5	4	8	30	5-4=1
S3	5	6	8	25	6-5=1
D	10	25	~~40~~ 20	75	

P1 ~~4-2=2~~ 4-3=1 8-1=7
5-2=3

	D1	D2	D3	Supply	P2
S1				~~20~~ 0	~~3-2=1~~
S2	5	4 $\boxed{25}$	8	~~30~~ 5	5-4=1
S3	5	6	8	25	6-5=1
D	10	~~25~~ 0	~~40~~ 20	75	

P2 5-5=0 ~~4-3=1~~ 8-8=0
6-4=2

	D1	D3	Supply	P3
S2	5 $\boxed{5}$	8	~~5~~ 0	8-5=3
S3	5	8	25	8-5=3
D	~~10~~ 5	20	75	

P3 5-5=0 8-8=0

	D1	D3	Supply	P4
S3	5 $\boxed{5}$	8 $\boxed{20}$	~~25~~ ~~20~~ 0	~~8-5=3~~
D	~~5~~ 0	~~20~~	75	

	D1	D2	D3	Supply
S1	2	3	1 $\boxed{20}$	20
S2	5 $\boxed{5}$	4 $\boxed{25}$	8	30
S3	5 $\boxed{5}$	6	8 $\boxed{20}$	25
D	10	25	40	75

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Optimal selection

$\Rightarrow 1\times20 + 5\times5 + 4\times25 + 5\times5 + 8\times20$

$\Rightarrow 330$

	D1	D2	D3	Supply	
S1	2	3	1 $\boxed{20}$	20	$u_1 = -7$
S2	5 $\boxed{5}$	4 $\boxed{25}$	8	30	$u_2 = 0$
S3	5 $\boxed{5}$	6	8 $\boxed{20}$	25	$u_3 = 0$
D	10	25	40	75	

$v_1 = 5 \quad v_2 = 4 \quad v_3 = 8$

$c_{ij} = u_i + v_j$

$c_{21} = u_2 + v_1$

$\Rightarrow 5 = 0 + v_1 \Rightarrow 5 = v_1$

$c_{22} = u_2 + v_2$

$\Rightarrow 4 = 0 + v_2 \Rightarrow 4 = v_2$

$c_{31} = u_3 + v_1$

$\Rightarrow 5 = u_3 + 5 \Rightarrow u_3 = 0$

$c_{33} = u_3 + v_3$

$\Rightarrow 8 = 0 + v_3 \Rightarrow 8 = v_3$

$c_{13} = u_1 + v_3$

$\Rightarrow 1 = u_1 + 8 \Rightarrow u_1 = -7$

$\Delta_{ij} = \cancel{c_{ij}}\ c_{ij} - (u_i + v_j)$

$\Rightarrow \Delta_{11} = c_{11} - (u_1 + v_1)$

$\Rightarrow \Delta_{11} = 2 - (-7 + 5)$

$\Rightarrow \quad \Rightarrow 2 + 2 \Rightarrow 4 = \Delta_{11}$

$\Delta_{12} = c_{12} - (u_1 + v_2)$

$\Rightarrow \Delta_{12} = 3 - (-7 + 4)$

$\Rightarrow \quad = 3 + 3 = 6 = \Delta_{12}$

$\Delta_{23} = c_{23} - (u_2 + v_3)$

$\Rightarrow \quad = 8 - (0 + 8)$

$\Rightarrow 8 - 8 = 0 = \Delta_{23}$

$\Delta_{32} = c_{32} - (u_3 + v_2)$

$\Rightarrow \quad = 6 - (0 + 4)$

$\Rightarrow 6 - 4 = 2 = \Delta_{32}$

Hence, all Δ_{ij} are +ve, the solution is optimal.